



Ph.D Entrance Examination

Subject: Computer Science

Syllabus

Digital Logic: Number system, Information representation, Computer arithmetic on fixed & floating point numbers, Boolean algebra, Combinational circuits, sequential circuits, Memory system, Processor organization, Input-output organization, pipe-line processing, static & dynamic interconnection networks.

Programming Languages: Paradigms, Data types, operations, Expressions, Control structures, I/O statements, Parameter parsing techniques. Language constructs for object-oriented, functional, logic & concurrent programming, Chomsky hierarchy of formal languages, finite automata & pushdown automata.

Optimization & Simulation Tools: Linear Programming: LPP in the standard form, canonical forms, conversion in standard form, Simplex prevention of cyclic computations in Simplex & Tableau, Big-M method, Dual Simplex & revised simplex.

Simulation: Analog vs. Digital simulation, Continuous & discrete system simulation, Simulation of Hypothetical Computer, Inventory system & Corporate system, simulation of PERT, Generation of uniform & Non- uniform random number, Monte Carlo method, Design of experiment, simulation languages.

Software Engineering: Development models, Metrics, Software Project Management, Analysis, Design: System design, detailed design, function-oriented, Object-oriented analysis & design, user interface design, Coding & Testing, Software quality & reliability, Object Modeling Technique (OMT) methodology.

Computer Graphics: Components of an Interactive Graphics system, Display systems, Input/output & storage devices, 2D geometry, Graphic operations, 3D Graphics, Animation, Graphic standards, Application Concepts, Projections and Hidden surface elimination.

Computer Networks: Fundamentals, Reference Models, Data Communication, Internetworking: Components and issues; Media access controls, virtual circuits & datagrams, Routing algorithm, Congestion control, Network Security, Firewalls, Internet architecture and protocols.

Data Base: Basic concepts, Characteristics of Database approach, Three-schema Architecture and Data Independence, Data Models, E-R Model, Relational Data Model, SQL Programming Techniques, Relational Database Design, Functional Dependencies, Normalization, Query Processing and Optimization, Transaction Processing Concepts, Concurrency Control Techniques and Recovery Techniques

Enhanced Data Models for Advanced Applications, Distributed Database and Client-Server Architectures.

Overview of Data Warehousing and OLAP, Data Mining Concepts

Emerging Database Technologies and Applications.

Data Structure: Arrays, String, Linked Lists - Singly, doubly & Circular List; Stacks, Queues, Priority Queues: Representation & Manipulation; Trees: Binary & Threaded Trees, traversal, Binary Search Tree, Huffman & AVL Trees, B Trees; Graphs: Adjacency Matrix, Path Matrix, Linked Representation, traversal; Searching & Sorting techniques..

Operating System: Functions, Multiprogramming, Multiprocessing & Multitasking, Memory Management, Virtual memory, Paging, Fragmentation. Concurrent Processing: Mutual exclusion, Critical regions, lock & unlock. Scheduling: CPU scheduling, I/O scheduling, Deadlock: avoidance & prevention; UNIX: Structure & commands of UNIX, Interfacing with UNIX, Editors & Compilers for UNIX, LEX & YACC, File system, System calls, Filters, Shell Programming.

System Programming: Assembly language fundamentals, Assemblers - 2 pass & single pass, Macros & Macro processors. Loading, Linking, relocation, program relocatability, linkage editing, Text editors, Programming Environments, Debuggers & program generation. Compilation & interpretation, Bootstrap compiler, Phase of compilation - lexical & syntax analysis, storage allocation, code optimization & generation.